

# Yoonseok Ra

Undergraduate Student  
Electrical and Computer Engineering  
Seoul National University

Email | Homepage | Google Scholar | GitHub | LinkedIn

## RESEARCH INTERESTS

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Control theory · Robotics · Stochastic control · Robust control · Model predictive control · Safe autonomous systems

## EDUCATION

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### Seoul National University (SNU), Seoul, Korea

*B.S. in Electrical and Computer Engineering, Mar. 2021 – Present (Expected Feb. 2028)*

- GPA of 4.3/4.3 in Spring 2026; 4.2/4.3 average over the last three semesters (2025–2026)
- Exchange student at EPFL (École Polytechnique Fédérale de Lausanne), Autumn 2026
- Military service as a KATUSA (Korean Augmentation to the U.S. Army), Jun. 2023 – Dec. 2024

### Daegu Science High School for the Gifted (DSHS), Daegu, Korea

*High School Diploma, Mar. 2018 – Feb. 2021*

## HONORS & AWARDS

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GPA-based Full-Tuition Merit Scholarship, Seoul National University, Fall 2025 – Spring 2026

5th Place, [2026 SNU FastMRI Challenge](#) (Team “경력직신입”)

## PUBLICATIONS

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*\*equal contribution — co-first authors*

Jaeuk Shin\*, **Yoonseok Ra\***, and Insoon Yang, “From Prediction Uncertainty to Conformalized Distance Fields for Safe Motion Planning,” submitted to *IEEE Transactions on Robotics (T-RO)*, 2026. [arXiv:2607.00776](#).

## RESEARCH EXPERIENCE

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### Conformal Distance Fields for Safe Motion Planning in Dynamic Environments

*Sep. 2025 – Present | Learning and Decision Systems Lab (LDS), SNU ECE*

- Co-developed a functional conformal prediction (FCP) framework that conformalizes the entire predicted distance field, yielding a distribution-free bound under which any feasible trajectory is certified safe
- Made per-step cost nearly independent of obstacle count by exploiting the low-rank residual field (offline functional-PCA envelope + lightweight online adaptive update)
- Embedded the bound as a tightened safety constraint in a sampling-based MPC (FCP-MPC) and validated it on the ETH-UCY pedestrian benchmarks and a dense 3D quadrotor task
- Co-first-authored manuscript submitted to IEEE T-RO, 2026 (see Publications)

## Evaluation of HU-based Data Clipping for Anatomically-Aware CT-to-MRI Conversion

*Jun. 2025 – Aug. 2025 | Laboratory for Imaging and Science Technology (LIST), SNU ECE*

- Investigated CT-to-MRI synthesis using conditional flow matching (CFM), a continuous-time generalization of diffusion models, focusing on HU-based data clipping as an anatomical prior
- Analyzed how physically motivated intensity constraints affect reconstruction stability and anatomical fidelity in generative models

## SELECTED TERM PROJECTS

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### Data-Calibrated Terminal Ingredients for Nominal MPC under Model Mismatch

*Spring 2026 | Term Project, Advanced Control Techniques, SNU ECE*

- Offline SDP-based calibration of MPC terminal ingredients that guarantees practical stability under model mismatch without modifying the online controller
- On a Duffing-like plant, cut terminal-time error  $\sim 10\times$  and recovered 210 stability cells where the nominal DARE design failed (none lost)

### A Lagrangian Field-Theoretic Interpretation of Discounted Markov Decision Processes

*Fall 2025 | Term Project, Graduate-level Course, SNU ECE*

- Developed a variational, field-theoretic formulation of discounted MDPs by analyzing the dual linear program and its continuous-time limit
- Showed that the Bellman flow constraint corresponds to a dissipative continuity equation, and that the Hamilton–Jacobi–Bellman equation emerges as an Euler–Lagrange condition

### RC Car Racing Challenge

*Fall 2025 | Introduction to Intelligent Systems, SNU ECE*

- Implemented hybrid PPO + rule-based controller deployed on a real RC car in ROS2

## SKILLS

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**Programming:** C, C++, Python (PyTorch, scikit-learn), MATLAB, Verilog, Java

**Frameworks & Tools:** ROS2

**Theoretical & Technical Background:** Control Theory, Optimal Control, Stochastic Control, Model Predictive Control, Convex Optimization, Semidefinite Programming, Conformal Prediction, Reinforcement Learning Theory, Probabilistic Safety, Algorithm Design

## SELECTED COURSEWORK

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Stochastic Control and Reinforcement Learning (Graduate-level, A+) · Mathematical and Numerical Optimization (A+) · Advanced Control Techniques (A+) · Introduction to Robotics (A+) · Fundamentals of Machine Learning (A+) · Fundamentals of Control Engineering (A+) · Signals and Systems (A+) · Introduction to Intelligent Systems (A+)

## ENGLISH PROFICIENCY

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TOEFL iBT 108/120 (Dec. 2025) · TEPS 492/600 (Feb. 2021) · TOEIC 945/990 (Mar. 2018)